

Engineering Statistics

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Chapter 7: Statistical Inference

Statistical inference

- ▶ **Statistical Inference** provides methods for drawing conclusions about a population from sample data.
 - ▶ **Confidence intervals**
 - ▶ estimating the value of a population parameter.
 - ▶ **Hypothesis testing**
 - ▶ assessing the evidence **against** a *claim* about a population.
- ▶ Statistical inferences are based on the **sampling distributions** of statistics.

Assumptions for statistical inference

- ▶ Simple **assumptions** for inference about a **population mean** μ :
- ▶ We have a good **simple random sample** (SRS) from the population of interest.
- ▶ The population is **Normally distributed** $N(\mu, \sigma^2)$.
- ▶ **We don't know μ , but we know σ .**

Confidence intervals (CI)

- ▶ Reporting only a point estimate is like fishing in a lake with a spear
 - ▶ We can throw a spear where we saw a fish but we will probably miss.
- ▶ If we report a point estimate, we probably won't hit the exact population parameter.
 - ▶ If you pick a random sample from a population, the observed sample mean, $\bar{x}$ would not actually equal to the true population mean, μ .

Confidence intervals (CI)

- ▶ If we toss a net in that area, we have a good chance of catching the fish.
- ▶ If we report a plausible range of values, we have a good shot at capturing the parameter.
 - ▶ A plausible range of values for the population parameter is called a **confidence interval**.

Confidence intervals (CI)

A **level $(1 - \alpha)100\%$ confidence interval** for a parameter:

$$\text{estimate} \pm \text{margin of error}$$

- ▶ The **estimate** is a **statistic**
- ▶ The **margin of error** represents the precision of our guess for the parameter.
 - ▶ It covers **random sampling error**.
- ▶ $(1 - \alpha)$ is called the confidence level.

CI for a Normal Population Mean μ (σ Known)

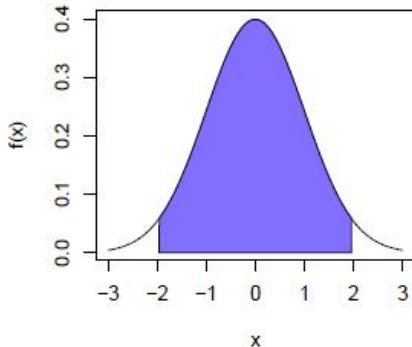
Let

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2).$$

- ▶ μ is unknown $\Rightarrow$ our interest.
- ▶ σ is known (*assumption*).

A level $(1 - \alpha)100\%$ **confidence interval** for the population mean μ is

$$\bar{x} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}$$



CI for a Normal population mean μ (σ known)

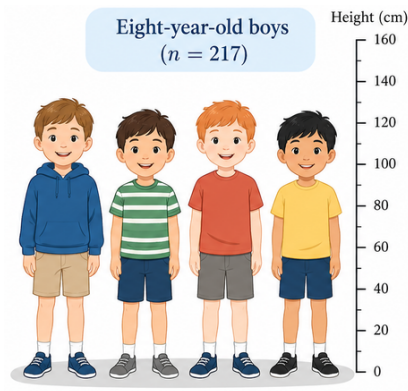
$$\bar{x} \pm z_{1-\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

- ▶ $\bar{x}$: **point estimate**
- ▶ $z_{1-\alpha/2} \times \frac{\sigma}{\sqrt{n}}$: the **margin of error**
 - ▶ $z_{1-\alpha/2}$ is called the critical value.
 - ▶ $z_{1-\alpha/2}$ can be found by
 - ▶ R function: `qnorm()`

Example: How tall are eight-year-old boys?

A national survey reports that the mean height of a sample of 217 eight-year-old boys was $\bar{x} = 132.5\text{cm}$ (52.2 in). Let's assume that

$$x_1, \dots, x_{217} \stackrel{iid}{\sim} N(\mu, 10^2).$$



Example: How tall are eight-year-old boys?

- ▶ $\bar{x} = (x_1 + \cdots + x_{217})/217 = 132.5.$
- ▶ $\frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{217}} \approx 0.68$
- ▶ With confidence level 95% ($1 - \alpha = 0.95$), $z_{1-\alpha/2} = 1.96$

A 95% **confidence interval** for the population mean μ :

$$132.5 \pm 1.96 \times 0.68 \iff (131.17, 133.83)$$

The length of confidence intervals

- ▶ $z_{1-\alpha/2}$ depends on the confidence level $(1 - \alpha)$
- ▶ The margin of error depends on $z_{1-\alpha/2}$
 - ▶ $(1 - \alpha) \uparrow \Rightarrow z_{1-\alpha/2} \uparrow \Rightarrow \text{margin of error} \uparrow$
 - ▶ $(1 - \alpha) \downarrow \Rightarrow z_{1-\alpha/2} \downarrow \Rightarrow \text{margin of error} \downarrow$
- ▶ The margin of error depends on sample size n
 - ▶ $n \uparrow \Rightarrow \frac{\sigma}{\sqrt{n}} \downarrow \Rightarrow \text{margin of error} \downarrow$
 - ▶ $n \downarrow \Rightarrow \frac{\sigma}{\sqrt{n}} \uparrow \Rightarrow \text{margin of error} \uparrow$

Example: How tall are eight-year-old boys?

A 95% **confidence interval** for the population mean μ :

$$132.5 \pm 1.96 \times 0.68 \iff (131.17, 133.83)$$

A 99% **confidence interval** for the population mean μ :

$$132.5 \pm 2.58 \times 0.68 \iff (130.75, 134.25)$$

- ▶ With confidence level 99% ($1 - \alpha = 0.99$), $z_{1-\alpha/2} = 2.58$

Interpretation of CIs

- ▶ The **confidence level** 95% gives the **proportion** that the interval will capture the true parameter value in **repeated samples**.
 - ▶ **The success rate of a method.**
- ▶ **NOTE:**
 - ▶ The confidence level 95% is NOT the probability that the calculated interval contains the unknown population parameter.

Interpretation of CIs

From the heights of eight-year-old boys example:

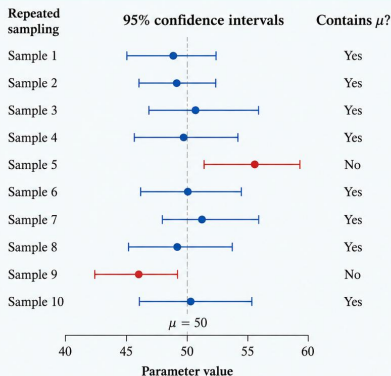
- ▶ A 95% **confidence interval** for the population mean μ :

$$132.5 \pm 1.96 \times 0.68 \iff (131.17, 133.83)$$

- ▶ "We got the numbers (131.17, 133.83) using a method that gives correct results 95% of the time"
- ▶ If we draw many random samples from the same population and compute 95% CIs, about 95% of the intervals would contain the true parameter μ .

Interpretation of CIs

Correct Interpretation of a 95% Confidence Interval



- A 95% confidence interval is a random interval computed from a random sample.
- If we **repeatedly** take samples of the same size and construct 95% confidence intervals in the same way, about 95% of those intervals will contain the true parameter μ .
- For any **single** interval, the parameter μ is either in the interval or not; we do not say there is a 95% probability that μ is in this specific interval.

Common misinterpretation (incorrect):

“There is a 95% probability that μ is in this interval.”

Correct interpretation:

“In repeated sampling, 95% of such intervals constructed by this method will contain μ .”

The confidence level refers to the long-run success rate of the method, not the probability that the parameter is in a particular interval.

Common misunderstandings of CI

Misunderstandings:

- ▶ The probability that the true parameter is in the computed confidence interval equals to 95%.
- ▶ 95% of the sample data lie within the confidence interval.
- ▶ There is a 95% probability of the parameter estimate from a repeat of the experiment falling within the confidence interval computed from a given experiment.

Using AI to Learn: Confidence Intervals

Ask AI

A sample of 217 eight-year-old boys has mean height 132.5 cm. Assume the population standard deviation is 10 cm. Construct a 95% confidence interval for the population mean and explain what the interval means.

What to Check

Does AI use the correct standard error and critical value?
Does it interpret the 95% confidence level as a repeated-sampling property rather than a probability about this particular interval?

Hypothesis testing

4 packs of cherry tomatoes are randomly sampled, *each labeled 1/2 lb (227g)*. The *average weight* from the 4 packs is 222g.



- ▶ Is the somewhat smaller weight simply due to chance variation?
- ▶ Is it evidence that the packing machine that sorts cherry tomatoes into packs needs revision?

Hypothesis testing

Someone makes a **claim** about the unknown value of a **true parameter**. We check whether or not this claim makes sense considering the “evidence” (sample).

- ▶ **Hypothesis testing:**

- ▶ tests a specific **hypothesis** using sample to decide on the validity of the hypothesis.

Null and alternative hypotheses

- ▶ The **null hypothesis** H_0 :
 - ▶ A very specific statement about a true parameter.
 - ▶ Usually, a statement of "**no effect**" or "**no difference**".
- ▶ The **alternative hypothesis** H_a :
 - ▶ A more general statement about the population that we are trying to find evidence for.

H_0 and H_a are statements about the **population**, NOT statements about samples.

Example: The weight of cherry tomato packs

4 packs of cherry tomatoes are randomly sampled, *each labeled 1/2 lb (227g)*. The *sample average weight* from the 4 packs is 222g.

- ▶ $H_0: \mu = 227g$
 - ▶ μ is the average weight of the population of packs
- ▶ $H_a: \mu \neq 227g$
 - ▶ μ is either larger or smaller than as stated in H_0

One-sided vs. two-sided alternatives

When $H_0 : \mu = \mu_0$ (a specific value),

- ▶ A **one-sided alternative**:

- ▶ $H_a : \mu < \mu_0$

- ▶ $H_a : \mu > \mu_0$

- ▶ A **two-sided alternative**:

- ▶ $H_a : \mu \neq \mu_0$

The choice of hypotheses: Statistical hypotheses should reflect our question based on **existing knowledge** on the topic, not some arbitrary choice.

- ▶ The hypotheses should express the hopes or suspicions we have **before** we see the data.

Example: Levels of inorganic phosphorus in the elderly

- ▶ Levels of inorganic phosphorus (P) in the blood are known to be normally distributed with mean 1.2 millimoles per liter (mmol/l) and standard deviation 0.1 mmol/l **among adults**.
- ▶ Since older people are more prone to bone fractures, our interest is to see **if phosphorus level decreases with age**.
 - ▶ $\bar{x} = 1.128$ from a sample of 12 people age 75 to 79.

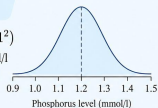
Example: Levels of inorganic phosphorus in the elderly

In adults

$$P \sim N(1.2, 0.1^2)$$

Mean = 1.2 mmol/l

SD = 0.1 mmol/l



Elderly sample (age 75 to 79)

Sample mean: $\bar{x} = 1.128$ mmol/l

Sample size: $n = 12$



A sample of 12 people age 75 to 79

Why study the elderly?

Older people are more prone to bone fractures, and lower phosphorus levels may be related to bone health.



Our question

Does the average phosphorus level decrease with age?

Our interest: seeking evidence for a **lower** phosphorus level in the elderly on average.

- ▶ $H_0 : \mu = 1.2$ (There is no difference between elderly and adult means)
- ▶ $H_a : \mu < 1.2$ (The elderly's mean is lower than adult's)

Example: Manufacturing aspirin tablets

Aspirin tablets are sold with the label "active ingredient: aspirin 325 mg." Since proper dosage is important, deviations from the expected dose distribution in either direction (too much or too little) is needed to be identified.

▶ $H_0 : \mu = 325$

▶ $H_a : \mu \neq 325$

Test statistics

- ▶ A **test statistic** calculated from the sample measures how far the data diverge from the null hypothesis H_0 .
 - ▶ We determine how many standard errors away from the null it is.
- ▶ Extreme values of the **test statistic** show that the data are far from what we would expect if H_0 is **true**.

Test statistics

Test	Test Statistics
One-sample Z -test	$T = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$
One-sample t -test	$T = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$
Matched pairs t -test	$T = \frac{\bar{X}_d - \mu_d}{s_d / \sqrt{n}}$
Two-sample t -test	$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

One-sample Z -test

- ▶ A hypothesis test for a population mean μ (σ known)
 - ▶ To test $H_0 : \mu = \mu_0$.
- ▶ Under $H_0 : \mu = \mu_0$, the sampling distribution of the sample mean is

$$\bar{X} \stackrel{H_0}{\sim} N\left(\mu_0, \frac{\sigma^2}{n}\right).$$

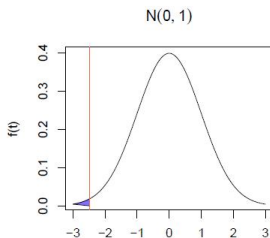
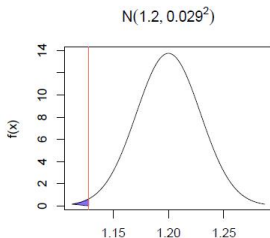
- ▶ Then, the test statistic is

$$T = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \stackrel{H_0}{\sim} N(0, 1).$$

Example: Levels of inorganic phosphorus in the elderly

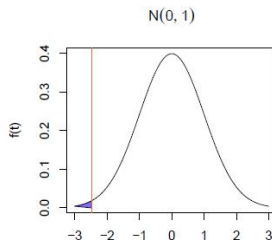
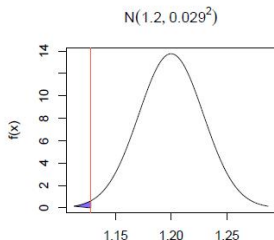
Under $H_0 : \mu = 1.2$, $X_1, X_2, \dots, X_{12} \stackrel{iid}{\sim} N(1.2, 0.1^2)$.

- ▶ The sample mean: $\bar{x} = 1.128$.
- ▶ Standard error: $\frac{\sigma}{\sqrt{n}} = \frac{0.1}{\sqrt{12}} = 0.029$.
- ▶ The observed test statistic $t = \frac{1.128 - 1.2}{0.029} = -2.48$.



P-value

Q. What is the probability of drawing a random sample such as ours, or even more extreme, **if H_0 is true?**



P-value

P-value:

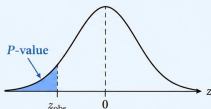
- ▶ Assuming H_0 is true, the probability that the test statistic would take a value as extreme as or more extreme than that actually observed **in the direction of H_a** .
 - ▶ $H_a : \mu > \mu_0 \implies P(T \geq \text{observed } t)$.
 - ▶ $H_a : \mu < \mu_0 \implies P(T \leq t)$.
 - ▶ $H_a : \mu \neq \mu_0 \implies 2 \times P(T \geq |t|)$.

Visualizing a P-Value

P-value: Area as Extreme as or More Extreme than the Observed Statistic

Left-Tailed Test

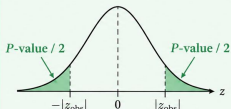
$$H_a: \mu < \mu_0$$



$$P\text{-value} = P(Z \leq z_{\text{obs}})$$

Two-Tailed Test

$$H_a: \mu \neq \mu_0$$



$$P\text{-value} = 2P(Z \geq |z_{\text{obs}}|)$$

Right-Tailed Test

$$H_a: \mu > \mu_0$$



$$P\text{-value} = P(Z \geq z_{\text{obs}})$$

- The P-value is the probability, assuming H_0 is true, of obtaining a test statistic as extreme as or more extreme than the observed value.
- A small P-value indicates stronger evidence against H_0 .

P-value = tail area as extreme as or more extreme than the observed statistic

Interpreting P-value

- ▶ Small P-values:
 - ▶ Strong evidence **against** H_0 .
 - ▶ We **can reject** H_0 .
 - ▶ The findings are "**statistically significant**."
- ▶ Large P-values:
 - ▶ Not enough evidence against H_0
 - ▶ We **fail to reject** H_0 .
 - ▶ It only means that the data are not inconsistent with H_0 .
 - ▶ It does NOT imply that H_0 is true.
 - ▶ Hence, we say "**cannot reject** H_0 " instead of "accept H_0 ".
 - ▶ A useful analogy is a trial: Failing to find sufficient evidence of guilt is not the same as proving innocence.

Significance level α

Q. How small a P-value?

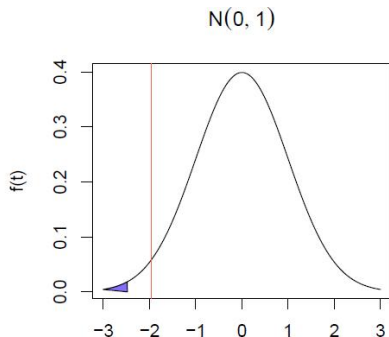
- ▶ $P(T \text{ has an extreme value} | H_0 \text{ is true}) \in [0, 1]$.
- ▶ We can compare it with some **fixed** values.

The significance level, α , is an arbitrary threshold that can be used statistical significance.

- ▶ α is decided **before** conducting the test.
- ▶ Common choice: $\alpha = 0.01$, or **0.05**.
- ▶ When $\text{P-value} \leq \alpha$, we can reject H_0 at level α .
- ▶ When $\text{P-value} > \alpha$, we fail to reject H_0 at level α .

Example: Levels of inorganic phosphorus in the elderly

- ▶ The observed test statistic $t = \frac{1.128 - 1.2}{0.029} = -2.48$.
- ▶ P-value = $P(T < -2.48) = 0.0066$.
- ▶ We can reject $H_0 : \mu = 1.2$ at level $\alpha = 0.05$.
 - ▶ $0.05 = P(Z < -1.645)$, $Z \sim N(0, 1)$.

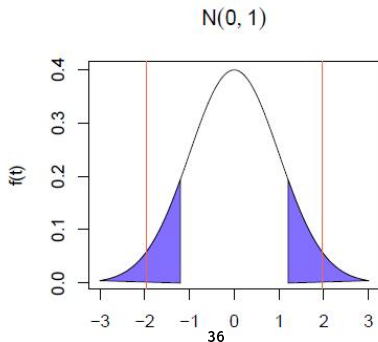


Example: Manufacturing aspirin tablets

- $H_0 : \mu = 325$ vs $H_a : \mu \neq 325$

Under H_0 , $X_1, \dots, X_{10} \overset{H_0}{\sim} N(325, 5^2)$ ($\sigma = 5$ known).

- ▶ $\bar{x} = 326.9, \sigma/\sqrt{n} = 1.58.$
- ▶ Observed $t = \frac{326.9 - 325}{1.58} = 1.203.$
- ▶ P-value $= 2 \times P(T > 1.203) = 0.229.$
- ▶ We fail to reject H_0 at level $\alpha = 0.05.$
 - ▶ $0.05 = 2 \times P(Z < -1.96), Z \sim N(0, 1).$



Using AI to Learn: Hypothesis Testing

Ask AI

A sample of 12 elderly adults has mean phosphorus level 1.128 mmol/l. Assume $\sigma = 0.1$. Test $H_0 : \mu = 1.2$ versus $H_a : \mu < 1.2$. Compute and interpret the P-value.

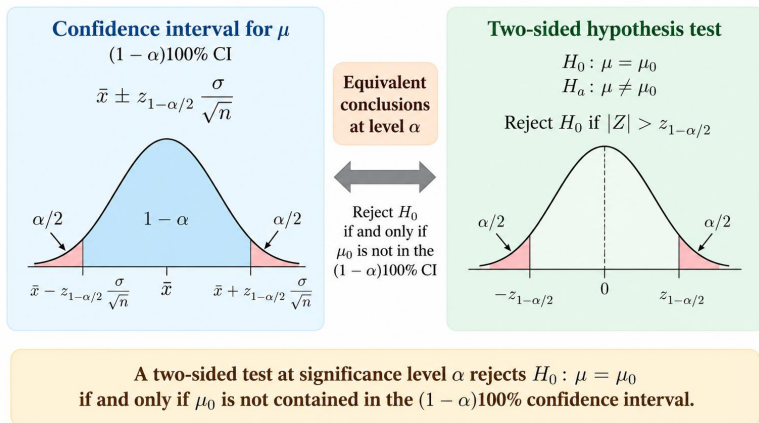
What to Check

Did AI use a left-tailed test? Did it compute the standard error and P-value correctly? Did it conclude “reject” or “fail to reject” H_0 , rather than claiming that a hypothesis has been proved?

Confidence interval and Hypothesis testing

Under H_0 , confidence intervals and two-sided tests are **equivalent**:

- ▶ When $(1 - \alpha)100\%$ CI **does not contain** μ_0 , we can reject H_0 .
- ▶ When $(1 - \alpha)100\%$ CI **contains** μ_0 , we fail to reject H_0 .



Example: Manufacturing aspirin tablets

Under $H_0 : \mu = 325$, $X_1, \dots, X_{10} \stackrel{H_0}{\sim} N(325, 5^2)$ ($\sigma = 5$ known).

- ▶ $\bar{x} = 326.9$, $\sigma/\sqrt{n} = 1.58$, $z_{1-\alpha/2} = z_{0.975} = 1.96$
- ▶ 95% CI: $326.9 \pm 1.96 \times 1.58 \iff (323.803, 329.997)$
- ▶ We fail to reject H_0 at level $\alpha = 0.05$.

Chapter Summary

- ▶ **Use samples to learn about populations.** Statistical inference uses sample information and sampling distributions to draw conclusions about population parameters.
- ▶ **Quantify uncertainty in estimation.** Confidence intervals give a range of plausible parameter values and account for random sampling variability.
- ▶ **Use hypothesis tests to evaluate claims.** Specify H_0 and H_a from the research question, then assess how consistent the observed data are with H_0 .
- ▶ **Interpret statistical evidence carefully.** Small P-values provide evidence against H_0 , but do not give the probability that H_0 is true. Failing to reject H_0 does not imply it is true.